

SIMULATION AND MODELING

ENCT353

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Course Objectives:

The objective of this course is to develop knowledge of modeling and simulation techniques for discrete and continuous systems. It emphasizes the development and analysis of simulation models, generation and testing of random numbers and variables, and application of simulation methods to evaluate the performance of stochastic systems

Chapter-1

Introduction to Simulation

Overview

- 1.1 System and system environment concepts
- 1.2 Continuous and discrete systems
- 1.3 Types of models
- 1.4 Model development life cycle
- 1.5 Simulation and steps in simulation study
- 1.6 Advantages and disadvantages of simulation
- 1.7 Monte-Carlo simulation
- 1.8 Discrete-event system simulation

1.1 System and system environment concepts

System

- The term system is derived from the Greek word **systema**, which means an organized relationship among functioning units or components.
- System exists because it is designed to achieve one or more objectives.
- We come into daily contact with the transportation system, the telephone system, the accounting system, the production system, and for two decades the computer system.
- There are more than a hundred definitions of the word system, but most seem to have a common thread that suggests that a **system is an orderly grouping of interdependent components linked together according to a plan to achieve a specific objective.**

1.1 System and system environment concepts

System

In a simple term **“A system is a collection of components or entities that interact together to achieve a specific goal.”**

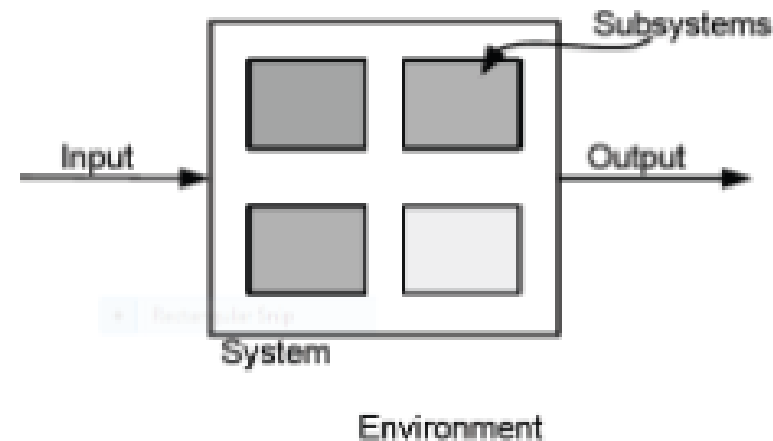
The study of the systems concepts, then, has three basic implications:

1. A system must be designed to achieve a predetermined objective.
2. Interrelationships and interdependence must exist among the components.
3. The objectives of the organization as a whole have a higher priority than the objectives of its subsystems.

1.1 System and system environment concepts

System

- Assembly of objects joined in some regular interaction or interdependence.
- A system exists and operates in time and space bounded inside system boundary



1.1 System and system environment concepts

System Components/ System Concepts – terms

- **Entity** = The object of interest of a system. An object that passes through the system. e.g. cars, buses in a traffic system
- **Attributes** = property/characteristics of an entity
- **Activity** = the process that causes change in the system
- **Event** = An instantaneous occurrence that may change the state of the system.
- **State** = collection of variables necessary to describe the system at any time.

- Examples

System	Entities	Attributes	Activities	Event
Traffic	Cars, Buses	Speed, Distance	Driving, Stopping	Arrival, Departure
Bank	Customers	Balance, Credit Status	Depositing Cash, Cash Withdrawal	Arrival, Departure
Communication	Message	Length, Priority	Transmitting Message	Message Breakdown
Supermarket	Customers	Shopping List	Clearance	Arrival, Departure

1.1 System and system environment concepts

- **System**
- Klir* gives a collection of 24 definitions one such definition is “ A system is a collection of components wherein individual components are constrained by connecting interrelationships such that the system as a whole fulfills some specific functions in response to varying demands”.
- Hence, system is a collection of entities that act and interact together toward the accomplishment of some logical end (computer, network, communication systems, queuing systems etc.)

*Klir, George J. , an approach to general systems theory, New York: Van Nostrand Reinhold Co,1969

1.1 System and system environment concepts

System Environment

- The **system environment** consists of external factors that influence the system but are not part of it.
- The changes occurring outside the system that affect the system is system environment.
- System may produce changes that do not react on the system but on the environment.
- A system is affected by the change in the environment.
- Example in a supermarket the factors controlling the arrival of the customers in the supermarket are outside the influence of supermarket and hence they are the part of environment.

Types of Activities

On the basis of place of Occurrence:

- **Endogenous Activities** – activities occurring within the system. Example: packaging, cheque clearing, etc.
- **Exogenous Activities** – activities occurring between the system and the environment. Example: Arriving customers

On the basis of its effect:

- **Deterministic Activities** – the outcome of the activity can be described completely in terms of its input. Example: switch off = no lights
- **Non- Deterministic (Astochastic) Activities** – the effect of the activity vary randomly over various possible outcomes. The random outcome can be measured and described in the form of probability distribution. Example: power failure

1.2 Continuous and discrete systems

- System is broadly categories into two types:

1. Continuous System

- System for which the state variables change continuously with respect to time.
- Continuous simulation is appropriate for systems with a continuous state that changes continuously over time.
- **Examples:**
 - **Temperature in a room:** Changes gradually based on heating/cooling
 - **Aircraft flight:** Altitude, speed, position change continuously
 - **Chemical reactions:** Concentration levels change smoothly
 - **Water tank drainage:** Water level decreases continuously
 - **Vehicle motion:** Speed and position change continuously

1.2 Continuous and discrete systems

2. Discrete System

- system for which the state variables change instantaneously at separated points in time
- Discrete simulation is appropriate for systems whose state is discrete and changes at particular time point and then remains in that state for some time.
- **Examples**
 - **Bank queue:** Customers arrive/depart at specific times; queue length jumps from 5 to 6
 - **Traffic light:** Changes from red → green → yellow at discrete moments
 - **Manufacturing:** Parts are completed at specific times; inventory counts change suddenly
 - **Computer network:** Packets arrive/depart at specific intervals

1.3 Types of models

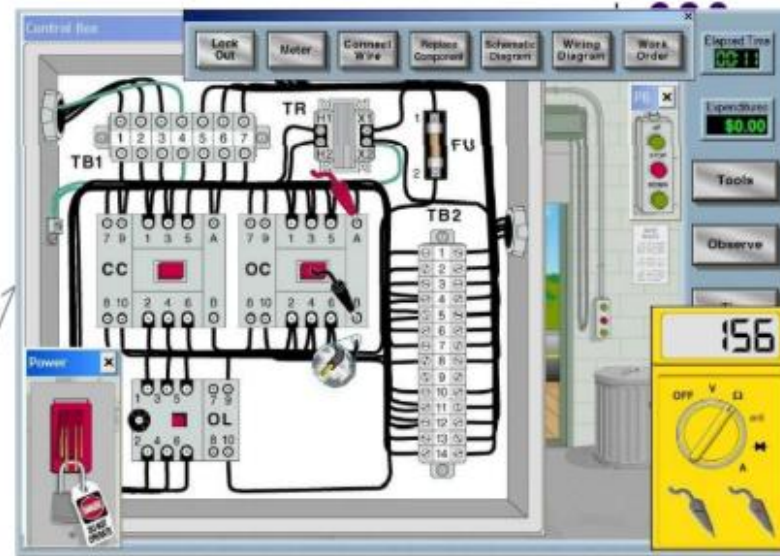
Model

- It is a simplification of the reality.
- A (usually miniature) representation of something; an example for imitation or emulation.
- A model is a representation of the construction and working of some system of interest.
- A model is similar to but simpler than the system it represents.
- One purpose of a model is to enable the analyst to predict the effect of changes to the system.
- A model is a simplified representation of a system at some particular point in time or space intended to promote understanding of the real system.

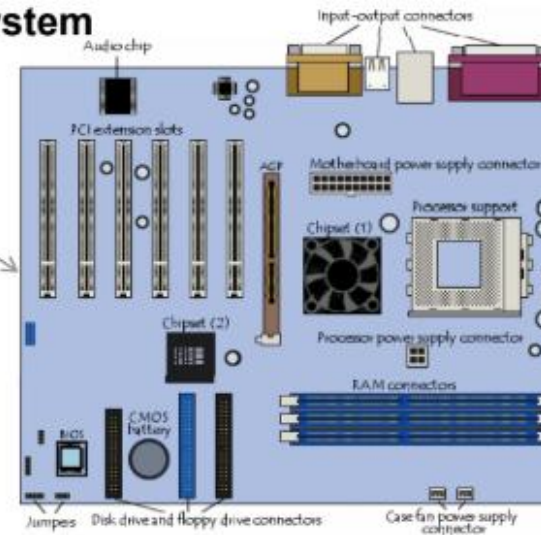
Examples

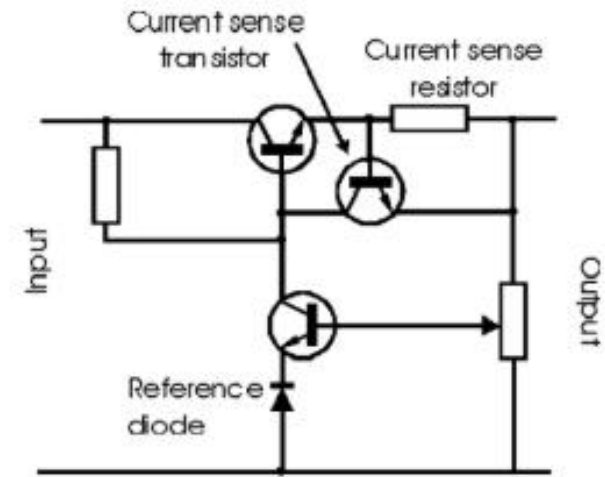


Real System (Motherboard)



Models of the system



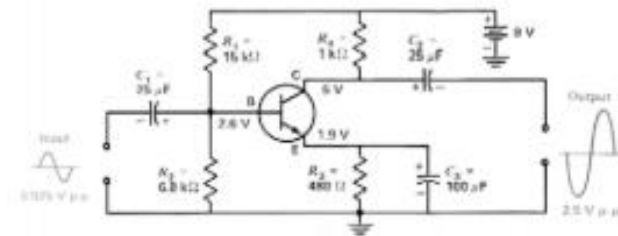


Models of the System

$$I = \frac{E}{R} \quad E = IR$$

$$R = \frac{E}{I} \quad P = EI$$

$$hfe = \frac{I_c}{I_b} \quad I_b = \frac{I_c}{hfe}$$



1.3 Types of models

System Modeling

- The model should be sufficiently detailed to permit valid conclusions to be drawn about the real system
- Modeling is the process of producing a model; On the one hand, a model should be a close approximation to the real system and incorporate most of its salient/relevant features.
- On the other hand, it should not be so complex that it is impossible to understand and experiment with it.
- A good model is a judicious tradeoff between realism and simplicity.

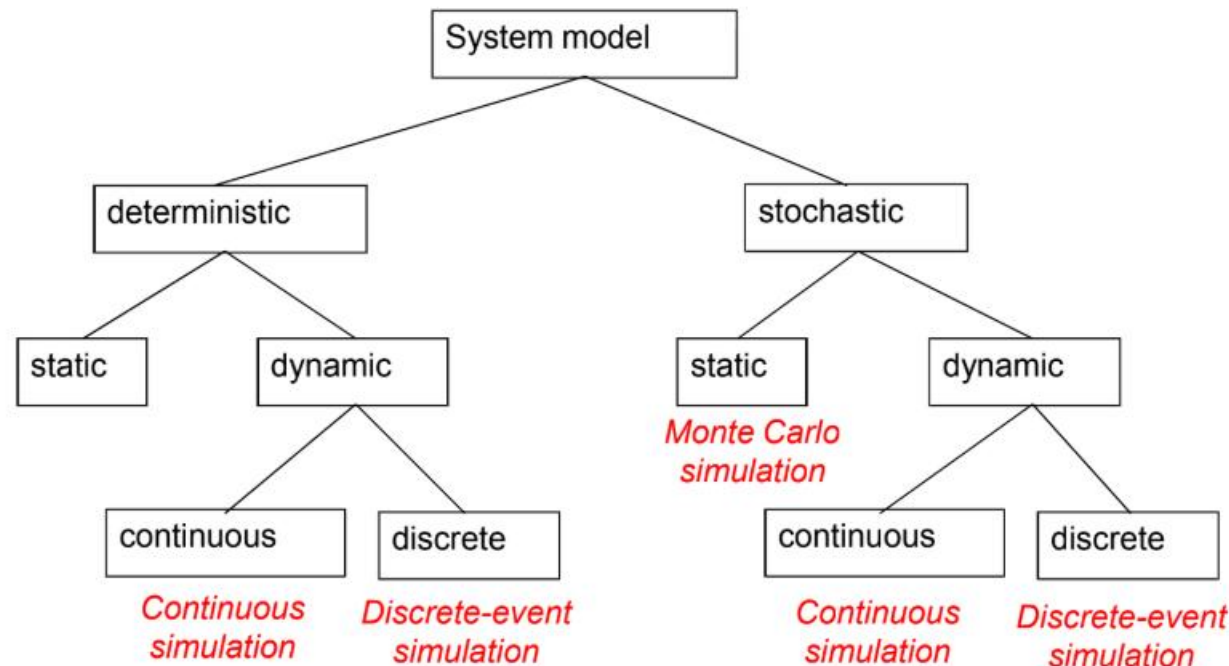
1.3 Types of models

System Modeling

- “Modeling is the use of mathematical techniques, specialized hardware and engineering judgment to create a representative stand for a real world environment, device, system or behavior.”
- Modeling of system could be categorized into 2 main tasks:
 1. Establishing model structure: determines system boundaries, identify attributes, entities and activities of system.
 2. Supplying the data: provide the values for the attributes and define relationship involved in the activities.

1.3 Types of models

On the basis of the nature of the system, Model of a system can be classified as:



1.3 Types of models

- Simulation models can be classified as being static or dynamic, deterministic or stochastic and discrete or continuous.
- **Deterministic models** have a known set of inputs, which result into unique set of outputs.
- In **stochastic model**, there are one or more random input variables, which lead to random outputs.
- A **static simulation** model represents a system, which does not change with time or represents the system at a particular point in time.
- **Dynamic simulation** models represent systems as they change over time.
- System in which the state of the system changes continuously with time are called **continuous systems** while the systems in which the state changes abruptly at discrete points in time called **discrete systems**.

1.3 Types of models

- **Stochastic vs. Deterministic**

Stochastic simulation

- There are one or more random input variables which leads to random outputs.
- a simulation that contains random (probabilistic) elements. Examples:- Inter-arrival time or service time of customers at a restaurant or store, Amount of time required to service a customer.
- Output is a random quantity (multiple runs required analyze output)

Deterministic simulation

- Models have a known set of inputs which result into unique set of outputs.
- a simulation containing no random elements. Examples:- Simulation of a digital circuit, Simulation of a chemical reaction based on differential equations
- Output is deterministic for a given set of inputs.

1.3 Types of models

- **Static vs. Dynamic Models**

Static models

- Model where time is not a significant variable.
- This model represents a system, which does not change with time or represents a system at a particular point in time.
- Examples:- Determine the probability of a winning solitaire hand

Dynamic models

- Represents a system that changes over time.
- Model focusing on the evolution of the system under investigation over time
- Example: Traffic simulation

1.3 Types of models

- **Continuous vs. Discrete**

Discrete

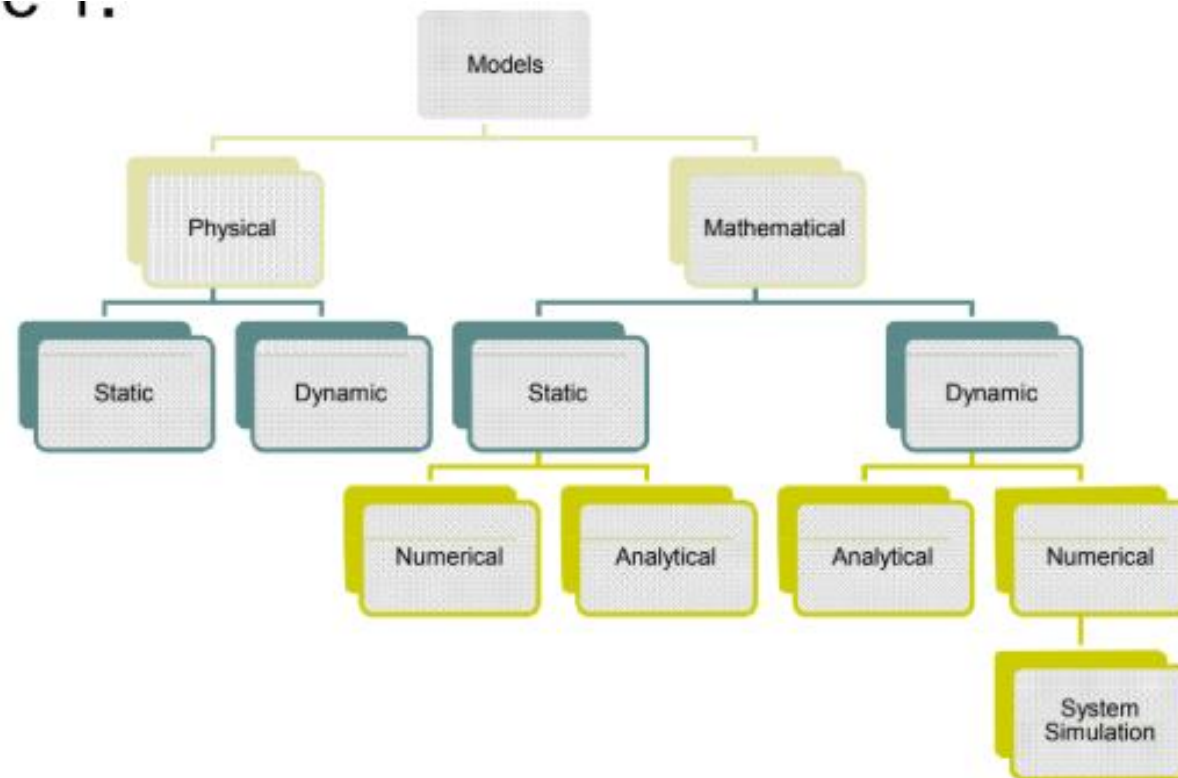
- Model represents the system in which state changes abruptly at discrete points of time.
- State of the system is viewed as changing at discrete points in time.
- An event is associated with each state transition.
- Events contain time stamp.

Continuous

- Model represents the system in which state changes continuously with time.
- State of the system is viewed as changing continuously across time.
- System typically described by a set of differential equations.

1.3 Types of models

- On the basis of the representation, Models used in system can be classified as:



1.3 Types of models

1. Physical Model

- Physical models are based on some analogy between such systems as mechanical and electrical or electrical and hydraulic.
- In a physical model of a system, the system attributes are represented by such measurements as a voltage or the position of a shaft.
- The system activities are reflected in the physical laws that drive the model.
- These models are known as iconic models.
- Example: Architect Modelling

2. Mathematical Models

- Mathematical models use symbolic notation and mathematical equation to represent a system.
- The system attributes are represented by variables, and the activities are represented by mathematical functions that interrelate the variables.

1.3 Types of models

3. Static Models

- These models can only show the values that system attributes take when the system is in balance i.e. representation of a system at a particular time or that may be used to represent the system in which time simply play no role.
- Static modeling includes class diagram and object diagrams and helps in depicting static constituents of the system.
- These models are rigid in nature as it is a time-independent view of the system.

1.3 Types of models

4. Dynamic Model

- These models represents the system that evolves over time which is resulted from system activity.
- It is the runtime model of the system.
- These models consists of sequence of operations, state changes, activities, interactions and memory.
- Dynamic models are flexible as it can be changed in real time.

1.3 Types of models

5. Numerical Model

- The model apply computational procedure to solve equations.
- Some analytical solution can become very complex requiring huge computing resources.
- In such case, the model must be studied using numerical model.
- Any assignment of numerical values that uses mathematical tables involves the numerical methods.

6. Analytical Model

- These model used the deductive reasoning of mathematical theory to solve a problem.
- Once we build mathematical model, it must then be examined to see how it can be used to answer the question of interest about the system it is supposed to represent.
- If a model is simple enough, it may be possible to work with its relationship and quantities to get an exact analytical solutions.
- In practice, only a certain form of equation can be solved analytically.
- Example: Boolean Equation

Differences between static modeling and dynamic modeling

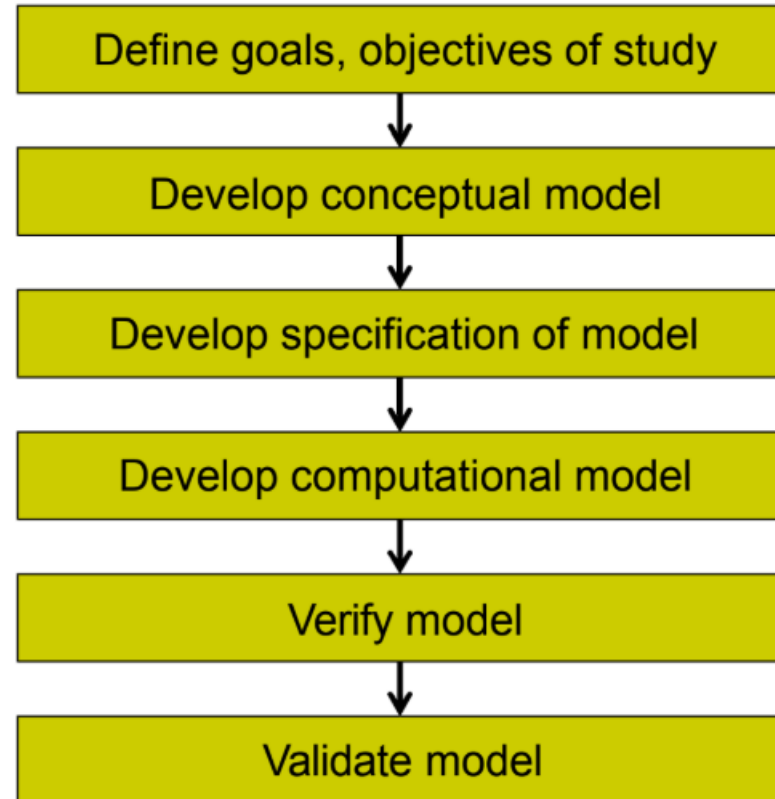
- The most notable difference between static and dynamic models of a system is that while a dynamic model refers to runtime model of the system, static model is the model of the system not during runtime.
- Another difference lies in the use of differential equations in dynamic model. Dynamic models keep changing with reference to time whereas static models are at equilibrium or in a steady state.
- Static modeling includes class diagram and object diagrams and help in depicting static constituents of the system.
- Dynamic modeling on the other hand consists of sequence of operations, state changes, activities, interactions and memory.
- Static modeling is more rigid than dynamic modeling as it is a time independent view of a system. It cannot be changed in real time and this is why it is referred to as static modeling.
- Dynamic modeling is flexible as it can be changed in real time.

1.4 Model development life cycle

The **Model Development Life Cycle** is the step-by-step process used to create a simulation model of a real system.

It helps us:

- understand the system,
- build the model,
- and test how the system behaves.



Fundamentally an iterative process

1.4 Model development life cycle

1. Define goals, objectives of Study

- Define the objective of simulation of the system of your choice.

2. Develop Conceptual Model

- Study the existing real system, its entities and attributes and develop its conceptual model.

3. Develop Specification of Model

- A more detailed information of the model including more specific attributes, required data, necessary algorithm, empirical data are collected.
- E.g. for traffic system: road geometry, signal timing, expected traffic demand, driver behavior, path planning for vehicles, etc.

1.4 Model development life cycle

4. Develop Computational Model

- Executable simulation model follows software approach using either general purpose programming language, special purpose simulation language or simulation package.
- While developing computational model, other non-functional requirement like performance, inter-operability with other tools/data are also considered.

5. Verification

- Deals with following question: “Did I build the model right?”
- Does the computational model match the specification model?
- Largely a software engineering activity (debugging)

1.4 Model development life cycle

6. Validation

- Did I build the right model?
- Does the computational model match the actual (or envisioned) system?
- Typically, compare against:
 - measurement of actual system
 - an analytical (mathematical) model of the system.
- By necessity always an incomplete activity.

1.5 Simulation and steps in simulation study

Simulation

- A simulation of a system is the operation of a model of the system.
- It is an experiment in a computer where the real system is replaced by the execution of the program.
- It is a program that mimics (imitate) the behavior of the real system.
- Simulation is a tool to evaluate the performance of a system, existing or proposed, under different configurations of interest and over long periods of real time.
- A simulation is the manipulation of a model in such a way that it operates on time or space to compress it, thus enabling one to perceive the interactions that would not otherwise be apparent because of their separation in time or space.

1.5 Simulation and steps in simulation study

Simulation

- Simulation is used before an existing system is altered or a new system built, to reduce the chances of failure to meet specifications, to eliminate unforeseen bottlenecks, to prevent under or over-utilization of resources, and to optimize system performance
- Can be used to study simple models but should not use it if an analytical solution is available. Real power of simulation is in studying complex models.

1.5 Simulation and steps in simulation study

Why Simulate?

- It may be too difficult, hazardous, or expensive to observe a real, operational system
- Parts of the system may not be observable (e.g., internals of a silicon chip or biological system)

Uses of simulations

- Analyze systems before they are built
- Reduce number of design mistakes
- Optimize design
- Analyze operational systems
- Create virtual environments for training, entertainment

1.5 Simulation and steps in simulation study

When to use Simulation?

Over the years tremendous developments have taken place in computing capabilities and in special purpose simulation languages, and in simulation methodologies. The use of simulation techniques has also become widespread. Following are some of the purposes for which simulation may be used.

1. Simulation is very useful for experiments with the internal interactions of a complex system, or of a subsystem within a complex system.
2. Simulation can be employed to experiment with new designs and policies, before implementing.
3. Simulation can be used to verify the results obtained by analytical methods and reinforce the analytical techniques.
4. Simulation is very useful in determining the influence of changes in input variables on the output of the system.
5. Simulation helps in suggesting modifications in the system under investigation for its optimal performance.
6. By simulating different capabilities for a machine, requirement can be determined.
7. By changing simulation inputs and observing the result outputs valuable information can be analyzed.

1.5 Simulation and steps in simulation study

Simulation not as an Appropriate Tool

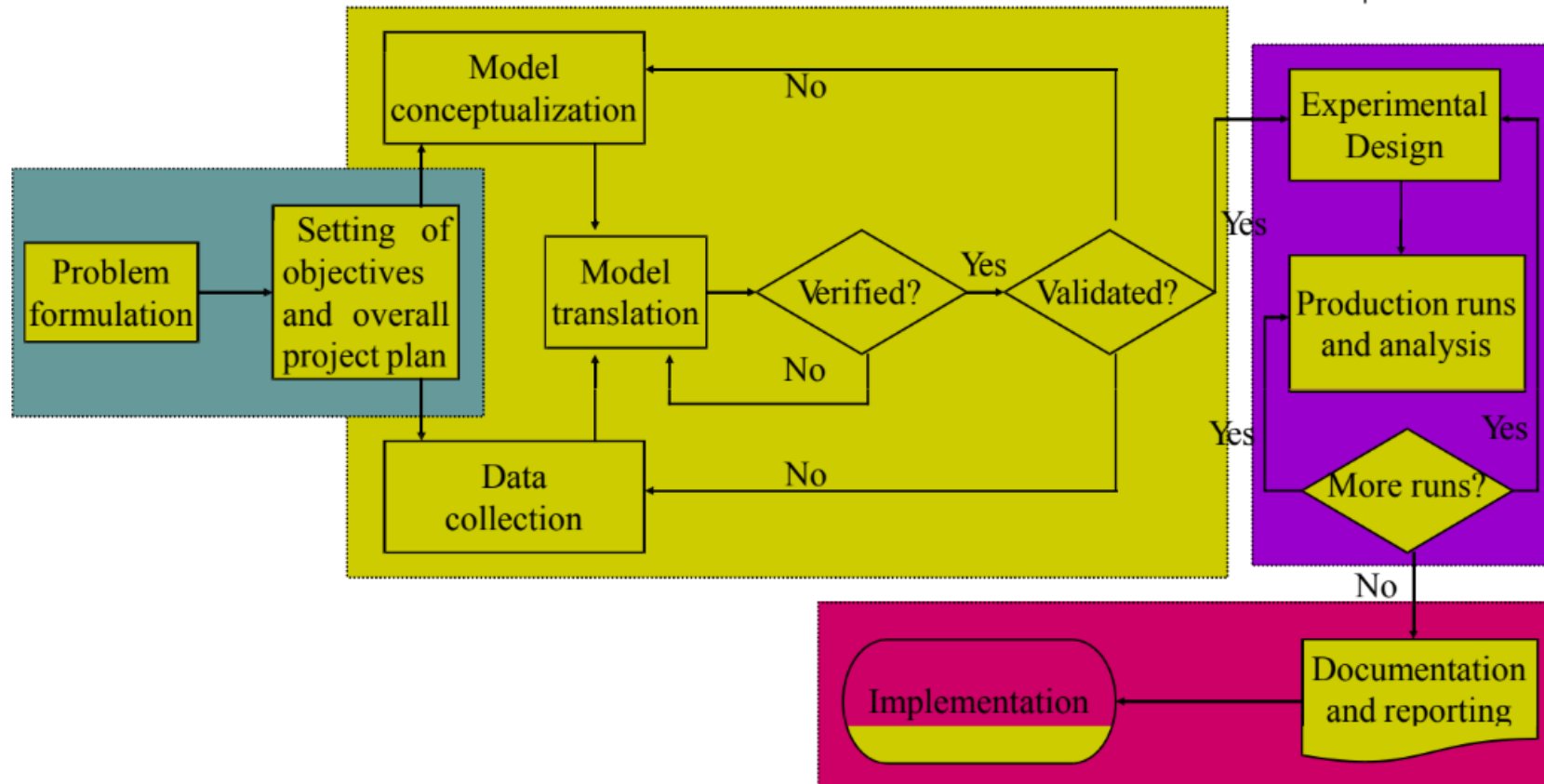
Simulation should not be used when the problem can be solve by using common sense. Example: Customer care

It should not be used:

- If the system problem can be solved analytically.
- If it is easier to perform direct experiment.
- If the resource or time is not available.
- If there is unavailability of verification and validation.
- If no data is available.

1.5 Simulation and steps in simulation study

Steps in simulation study



1.5 Simulation and steps in simulation study

Simulation is a very general method of studying problems. No formal procedure can be given for showing how a simulation study will proceed. Above figure shows the general steps involved.

1. Problem Formulation

- Initial step is to describe the problem to be solved. It should be a clear statement of what questions are being asked and what measurements need to be taken in order to answer these questions.

2. Setting of Objectives

- The objectives indicate the questions to be answered by simulation. Assuming that the simulation is appropriate, the overall project plan should include a statement of the alternative system to be considered, the cost of study, number of days required, etc.

3. Model Conceptualization

- It is best to start with a simple model and build towards greater complexity. The art of modeling is enhanced by an ability to abstract the essential features of a problem, to select and modify basic assumptions that characterize the system and then to enrich and elaborate the model until an useful approximations results.

1.5 Simulation and steps in simulation study

4. Data Collection

- Begins usually together with the early stages of model building. As the complexity of model changes, the required data elements can also change.

5. Model Translation

- Most real world system result in models that require a great deal of information storage and computation, so the model must be entered into a computer recognizable format. The modeler must decide whether to program the model in a simulation language or to use special-purpose simulation software.

6. Verified?

- Is this computer program performing properly? With complex models, it is difficult to translate a model successfully without a good deal of debugging. If the input parameters and logical structure of the model are correctly represented, verification has been completed.

1.5 Simulation and steps in simulation study

7. Validated?

- It is usually achieved through the calibration of the model.

8. Experimental Design

- The alternatives that are to be simulated must be determined. The decision concerning which alternatives to simulate will be a function of runs that have been completed and analyzed.

9. Production Runs and Analysis

- Production runs and their subsequent analysis are used to estimate measures of performance for the system design that are being simulated.

1.5 Simulation and steps in simulation study

10. More Runs?

- Given the analysis of runs that have been completed, the analyst determines whether additional runs are needed and what design these additional experiments should follow.

11. Documentation and Reporting

- There are two types of documentation: Program and Progress.
- Program documentation is necessary for numerous reasons. It helps to make decisions and helps to modify the program.
- Progress report provides important written history of a simulation project.
- The final report should include: a model specification, prototype demonstrations, animations, program documentation, progress report and presentations.

1.6 Advantages and disadvantages of simulation

Advantages of Simulation

- Simulation helps to learn about real system, without having the system at all. For example the wind tunnel testing of the model of an aeroplane does not require a full sized plane.
- In many situations experimenting with actual system may not be possible at all. For example, it is not possible to conduct experiment, to study the behavior of a man on the surface of moon. In some other situations, even if experimentation is possible, it may be too costly and risky,
- In the real system, the changes we want to study may take place too slowly or too fast to be observed conveniently. Computer simulation can compress the performance of a system over years into a few minutes of computer running time.

1.6 Advantages and disadvantages of simulation

Advantages of Simulation

- Conversely, in systems like nuclear reactors where millions of events take place per second, simulation can expand the time to required level.
- Through simulation, management can foresee the difficulties and bottlenecks, which may come up due to the introduction of new machines, equipment and processes. It thus eliminates the need of costly trial and error method of trying out the new concepts.
- Simulation Models are comparatively flexible and can be modified to accommodate the changing environment to the real situation.

1.6 Advantages and disadvantages of simulation

Advantages of Simulation

- Extensive computer software packages are available, making it very convenient to use fairly sophisticated simulation models.
- Simulation is a very good tool of training and has advantageously been used for training the operating and managerial staff in the operation of complex system. Space engineers simulate space flights in laboratories to train the future astronauts for working in weightless environment.
- Airline pilots are given extensive training on flight simulators, before they are allowed to handle real planes.

1.6 Advantages and disadvantages of simulation

Disadvantages of Simulation

- Model building requires special training. It is an art that is learned over time and through experience. Furthermore, if two models are constructed by two competent individuals, they may have similarities, but it is highly unlikely that they will be the same.
- Simulation results may be difficult to interpret. Since most simulation outputs are essentially random variables, it may be hard to determine whether an observation is a result of system interrelations or randomness.
- Simulation is used in some cases when an analytical solution is possible, or even preferable.
- Simulation modeling and analysis can be time consuming and expensive.

Limitations of Simulation

- Simulation generates a way of evaluating solutions but does not generate solution themselves.
- Simulation is not precise. It does not yield an answer but merely provides a set of the system's responses to different operating conditions. In many cases, this lack of precision is difficult to measure.
- It is a trial and error method that may produce different solutions in repeated runs.
- A good simulation model may be very expensive. Simulation often requires a significant amount of computer time and is therefore expensive.

Areas of Application

- **Manufacturing:** Design analysis and optimization of production system, materials management, capacity planning, layout planning, and performance evaluation, evaluation of process quality.
- **Business:** Market analysis, prediction of consumer behavior, and optimization of marketing strategy and logistics, comparative evaluation of marketing campaigns.
- **Military:** Testing of alternative combat strategies, air operations, sea operations, simulated war exercises, practicing ordinance effectiveness, inventory management.
- **Healthcare applications:** such as planning of health services, expected patient density, facilities requirement, hospital staffing , estimating the effectiveness of a health care program.

Areas of Application

- **Communication Applications:** Such as network design, and optimization, evaluating network reliability, manpower planning, sizing of message buffers.
- **Computer Applications:** Such as designing hardware configurations and operating system protocols, sharing networking. Client/Server system architecture
- **Economic applications:** such as portfolio management, forecasting impact of Govt. Policies and international market fluctuations on the economy. Budgeting and forecasting market fluctuations.
- **Transportation applications:** Design and testing of alternative transportation policies, transportation networks-roads, railways, airways etc. Evaluation of timetables, traffic planning.

Areas of Application

- **Environment application:** Solid waste management, performance evaluation of environmental programs, valuation of pollution control systems.
- **Biological applications:** Such as population genetics and spread of epidemics.
- **Business process Re-engineering:** Integrating business process re-engineering with image –based work flow, using process modeling and analysis tool.

1.7 Monte-Carlo simulation

- Simulation can also be defined as a technique of performing sampling experiments on the model of the system.
- This is called stochastic simulation and is a part of simulation techniques.
- Because sampling from a particular probability distribution involves the use of random numbers, stochastic simulation is sometimes called Monte Carlo Simulation
- **Monte Carlo simulation** is a computational technique that uses random sampling to solve problems and estimate numerical results.
- It works by running many random simulations to approximate the behavior of a system when analytical solutions are difficult or impossible.

1.7 Monte-Carlo simulation

How Monte Carlo Simulation Works?

1. Define the Problem

- Identify variables with uncertainty
- Determine their probability distributions

2. Generate Random Values

- Create random samples from each variable's distribution
- Use random number generators

3. Run Simulation

- Calculate outcomes using the random values
- Repeat thousands or millions of times
- Each iteration uses a different random sample

4. Analyze Results

- Collect all outcomes
- Calculate statistics (average, standard deviation, percentiles)
- Visualize distribution of results

1.7 Monte-Carlo simulation

Simple Example: Estimating the value of π

Imagine:

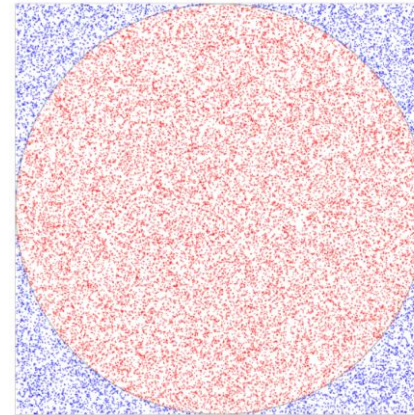
- A square with side length = 2
- A circle inside the square with radius = 1
- The circle fits perfectly inside the square.

Areas

- Area of square = $2 \times 2 = 4$
- Area of circle = $\pi r^2 = \pi$

So, $\frac{\text{Area of circle}}{\text{Area of square}} = \frac{\pi}{4}$

$$\pi = 4 \times \frac{\text{Points inside circle}}{\text{Total points}}$$



1.7 Monte-Carlo simulation

Simple Example: Estimating the value of π

Algorithm

1. Generate a random point (x, y) inside a square of side 2 centered at the origin.
2. Determine whether the point falls inside the unit circle inscribed in the square by checking whether $x^2 + y^2 \leq 1$.
3. Repeat steps 1 and 2 for a large number of points (e.g., 10^7).
4. Calculate the ratio of the number of points that fell inside the circle to the total number of points generated.
5. Multiply the ratio by 4 to estimate the value of π .

1.7 Monte-Carlo simulation

Simple Example: Estimating the value of π

Results (Assumptions)

10 points: $\pi \approx 3.2$

100 points: $\pi \approx 3.16$

1,000 points: $\pi \approx 3.142$

10,000 points: $\pi \approx 3.1416$

1,000,000 points: $\pi \approx 3.141592...$

2. Disadvantages of Monte Carlo Simulation

1. Computationally Expensive: A large number of simulation runs may require significant computational time and resources, especially for complex models.
2. Accuracy Depends on Input Data: Incorrect assumptions or inaccurate input probability distributions can lead to misleading results.
3. No Guaranteed Exact Solution: Monte Carlo simulation provides approximate results rather than an exact mathematical solution.
4. Requires Expertise: Developing an appropriate simulation model requires knowledge of probability, statistics, and the problem domain.
5. Random Variation: Different simulation runs may produce different results because of the use of random numbers.
6. Difficult to Validate: It may be challenging to determine whether a simulation model accurately represents the real-world system.
7. Garbage In, Garbage Out: If the input data or model assumptions are wrong, the output will also be unreliable, regardless of the number of simulation runs.
8. Interpretation Can Be Difficult: Understanding and interpreting probability distributions and simulation results may be challenging for non-technical users.
9. Time-Consuming Model Development: Building, testing, and refining a realistic simulation model can take considerable time.
10. Rare Events Require More Runs: Estimating the probability of very rare events may require an extremely large number of simulation trials.

1.7 Monte-Carlo simulation

Assignment

Q. Advantages and Disadvantages of Monte Carlo simulation

Q. Applications of Monte Carlo Simulation

The major advantages are:

1. Handles Uncertainty: It can model problems involving uncertain variables such as demand, cost, time, and failure rates.
2. Applicable to Complex Problems: It can solve complex mathematical and real-world problems that are difficult to solve using analytical methods.
3. Provides Multiple Outcomes: Instead of producing a single result, it generates a range of possible outcomes along with their probabilities.
4. Risk Analysis: It helps identify and evaluate risks by estimating the probability of different outcomes.
5. Flexibility: It can be applied to various fields, including engineering, finance, healthcare, business, and science.
6. What-if Analysis: It allows decision-makers to examine how changes in input variables affect the final result.
7. Cost-Effective: It can reduce the need for expensive physical experiments, prototypes, and real-world testing.
8. Supports Decision-Making: It provides quantitative information that helps managers and engineers make informed decisions under uncertainty.
9. Easy to Automate: With computers, a large number of simulation trials can be performed quickly.
10. Sensitivity Analysis: It helps identify which input variables have the greatest influence on the output.

1.7 Monte-Carlo simulation

Q. A bakery keeps records of a popular brand of cake:

Daily Demand	0	15	25	35	45	50
Probability	0.01	0.15	0.20	0.50	0.12	0.02

Given set of Random Numbers:

48,78,09,51,56,77,15,14,68 and 09

- i) Using these sequences, simulate demand for next 10 days.
- ii) Also find stock situation if owner made 35 cake/day.
- iii) Also find average daily demand on the basis of simulated data.

Steps:

1. Establish probability distribution
2. Cumulative Probability Distribution
3. Setting Random Number Interval
4. Generating Random Number
5. Finding the answer of the question

1. Establish probability distribution
2. Cumulative Probability Distribution
3. Setting Random Number Interval

Daily Demand	Probability	Cumulative Probability	Random Number Interval
0	0.01	0.01	0-0
15	0.15	0.16	01-15
25	0.20	0.36	16-35
35	0.50	0.86	36-85
45	0.12	0.98	86-97
50	0.02	1.00	98-99

Generating Random Number

48,78,09,51,56,77,15,14,68,09

Day	Random Number	Simulated Demand	Stock
1	48	35	0
2	78	35	0
3	09	15	20
4	51	35	0
5	56	35	0
6	77	35	0
7	15	15	20
8	14	15	20
9	68	35	0
10	09	15	20
		Total Demand=270	Total Stock= 80

Average daily demand = $270/10 = 27$

Q. Strong is a dentist who schedules all her patients for 30 minutes appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, and the time actually needed to complete the work:-

Category	Time Required	No. of Patients
Filling	45min	40
Crown	60min	15
Cleaning	15min	15
Extracting	45min	10
Checkup	15min	20

Use monte carlo simulation to find average waiting time of patients as well as idle time of doctor if clinic operates for 4 hours. Assume all patients arrive at clinic on scheduled time with arrival time starting at 8:00 am.

Use following random numbers

40,82,11,34,25,66,17,79

Category	Time Required	No. of Patients	Probability Distribution		
Filling	45min	40	0.4		
Crown	60min	15	0.15		
Cleaning	15min	15	0.15		
Extracting	45min	10	0.1		
Checkup	15min	20	0.2		

Category	Time Required	No. of Patients	Probability Distribution	Cumulative Probability	
Filling	45min	40	0.4	0.4	
Crown	60min	15	0.15	0.55	
Cleaning	15min	15	0.15	0.70	
Extracting	45min	10	0.1	0.80	
Checkup	15min	20	0.2	1.00	

Category	Time Required	No. of Patients	Probability Distribution	Cumulative Probability	Random Number Interval
Filling	45min	40	0.40	0.40	0-39
Crown	60min	15	0.15	0.55	40-54
Cleaning	15min	15	0.15	0.70	55-69
Extracting	45min	10	0.10	0.80	70-79
Checkup	15min	20	0.20	1.00	80-99

Patient	Scheduled Arrival	Random Number	Category	Service Time Needed
1	8:00	40	Crown	60min
2	8:30	82	Checkup	15min
3	9:00	11	Filling	45min
4	9:30	34	Filling	45min
5	10:00	25	Filling	45min
6	10:30	66	Cleaning	15min
7	11:00	17	Filling	45min
8	11:30	79	Extracting	45min

Patient	Scheduled Arrival	Service Start	Service duration	Service Ends	Waiting time(mins)	Idle time
1	8:00	8:00	60min	9:00	0	0
2	8:30	9:00	15min	9:15	30	0
3	9:00	9:15	45min	10:00	15	0
4	9:30	10:00	45min	10:45	30	0
5	10:00	10:45	45min	11:30	45	0
6	10:30	11:30	15min	11:45	60	0
7	11:00	11:45	45min	12:30	45	0
8	11:30	12:30	45min	1:15	60	0
					Total =285	

Average waiting time = $285/8 = 35$

Q. An air force station has a schedule of 15 transport flights leaving per day, each with one pilot. Three reserve pilots are available to replace any pilot who falls sick. The probability distribution for the daily number of pilots who fall sick is as follows:

Number of pilots sick	0	1	2	3	4	5
Probability	0.15	0.20	0.25	0.15	0.15	0.10

Use the Monte Carlo simulation to estimate the utilization of reserved pilots. What is the probability of canceling one flight due to non-availability of pilot? Simulate the system for 5 days.